Transition Rates and Statistical Mechanics of Superconducting Qubits

Onri Jay Benally

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Abstract

Transition rates connect microscopic energy exchange in a superconducting circuit to the lifetimes and coherence times reported as hardware specifications. Because surrounding materials and wiring both exchange energy with the quantized circuit motion and perturb its frequency, a physical description must account for population dynamics together with phase coherence. Starting from a Josephson circuit Hamiltonian, this report develops that description through thermal statistical mechanics, quantum noise spectra, and the density matrix master equation. The resulting population recovery rate is the sum of downward and upward transition rates, while exponential coherence decay contains half that recovery rate together with pure dephasing. A consistent spectral convention identifies the direction of energy transfer and links thermal detailed balance to the fluctuation dissipation theorem, allowing dielectric participation, resonator coupling, and quasiparticle tunneling to enter quantitative loss models. Worked calculations for a 5 GHz device connect thermal populations and decay probabilities to gate error scales and electromagnetic design targets, while statistical estimation relates repeated measurements to fitted rates and their uncertainty. By linking each specification to its defining experiment, the report provides a mathematical reference for understanding coherence as an engineering property of an open physical circuit and for explaining that connection in technical presentations.

1 The physical circuit and its quantum states

A superconducting qubit encodes a quantum bit in the lowest two energy eigenstates of a fabricated electrical circuit. While capacitance and inductance store electric and magnetic or kinetic energy, respectively, a Josephson element supplies the nonlinearity required to address one transition independently of its neighbors. These components define a collective electrical degree of freedom, which in a transmon is the superconducting phase difference across a Josephson junction and its conjugate charge (Koch et al., 2007).

For a single transmon mode, the charging and Josephson contributions combine in the Hamiltonian

$$\widehat{H}_q = 4E_C\left(\hat{n} - n_g\right)^2 - E_J \cos\hat{\varphi}, \left[\hat{\varphi}, \hat{n}\right) = i. \tag{1}$$

The Cooper pair number operator n̂ and the dimensionless offset charge n_g enter the charging term, whose energy scale is $E_C = \frac{e^2}{2C_\Sigma}$; the Josephson contribution instead has energy $E_J = \frac{\hbar I_c}{2e}$. In these definitions, C_Σ is the total shunting capacitance, I_c is the critical current, e is the elementary charge, and $\hbar = \frac{h}{2\pi}$. Because the phase dependent Josephson energy makes neighboring level spacings unequal, diagonalizing equation (1) gives eigenstates |j⟩ and energies E_j that define distinct transition frequencies:

$$\widehat{H}_q \left| j \right\rangle = E_j \left) j \right\rangle, \omega_{ij} = \frac{E_j - E_i}{\hbar}, f_{ij} = \frac{\omega_{ij}}{2\pi}. \tag{2}$$

In the transmon regime $E_J / E_C \gg 1$, the first transition frequency and angular anharmonicity follow the approximations

$$\omega_{01} \simeq \frac{\sqrt{8E_J E_C} - E_C}{\hbar}, \alpha \equiv \omega_{12} - \omega_{01} \simeq -\frac{E_C}{\hbar}. \tag{3}$$

Anharmonicity separates the desired |0⟩ ↔ |1⟩ transition from |1⟩ ↔ |2⟩, allowing a microwave field resonant with the first transition to drive a controlled coherent rotation. Environmental fields couple through the same electrical coordinates, where they drive unwanted transitions or perturb the transition frequency; understanding the hardware therefore requires both the circuit spectrum and the spectrum of the fields that reach it.

Projecting onto the two computational states and removing a constant energy gives the effective circuit Hamiltonian

$$\widehat{H}_q = \frac{\hbar\omega_{01}}{2}\hat{\sigma}_z, \hat{\sigma}_z = \left|1\right\rangle\left\langle 1\right) - \left|0\right\rangle\left\langle 0\right). \tag{4}$$

The energy splitting ΔE = ħω₀₁ = hf₀₁ connects the quantum levels to their spectroscopic frequency. Throughout this report, angular frequency parameters such as ω, g, and κ are expressed in radians per second, ordinary frequencies such as f are expressed in hertz, and transition rates are expressed in inverse seconds. Keeping these conventions explicit prevents an unintended factor of 2π from entering a lifetime or coupling calculation.

Including the environmental degrees of freedom gives the total Hamiltonian $\widehat{H}=\widehat{H}_q+\widehat{H}_B+\widehat{H}_{int}$, where $\widehat{H}_B$ describes the bath. Tracing over that bath produces the reduced qubit state $\rho=Tr_B\rho_{total}$, whose diagonal elements give state populations and whose off diagonal elements hold the phase information required for interference. This reduced description connects the materials and microwave environment to the quantum information stored in the circuit (Blais et al., 2021).

2 A transition rate is a conditional probability per unit time

A downward transition rate $\Gamma\downarrow$ specifies how rapidly an occupied excited state transfers population into the ground state, with the corresponding upward rate $\Gamma\uparrow$ defined through the short interval probabilities

$$Pr(1\to 0\,during\,dt|1)=\Gamma_{\downarrow}dt+O(dt^2),Pr(0\to 1\,during\,dt|0)=\Gamma_{\uparrow}dt+O(dt^2). \tag{5}$$

Because a downward transition requires occupation of $|1\rangle$, $\Gamma\downarrow$ specifies a conditional probability per unit time, making it distinct from the unconditional fraction of shots reported in an experiment. This conditional rate is the hazard for leaving the occupied excited state, so constant rates and memoryless dynamics give the survival probability and first downward jump statistics

$$P_{surv}(t)=e^{-\Gamma_{\downarrow}t},w_{\downarrow}(t)=\Gamma_{\downarrow}e^{-\Gamma_{\downarrow}t},\langle t_{wait}\rangle=\Gamma_{\downarrow}^{-1}. \tag{6}$$

The first jump waiting time density $w\downarrow$ describes an exponential distribution in which the individual jump time is random but the ensemble statistics are fixed. When upward transitions are negligible, the excited population equals the survival probability, so the number K still excited at delay t among N independently prepared qubits, or N repeated preparations of one qubit, follows the binomial distribution

$$K\sim Binomial(N,p),p=e^{-\Gamma_{\downarrow}t},E[K]=N\,p,Var(K)=N\,p(1-p). \tag{7}$$

An exponential decay curve therefore describes the expected population across repeated preparations. At finite temperature, subsequent excitation after a downward jump makes the final population depend on both transition directions, which gives the rate equation for $p_1 = \rho_{11}$

$$\frac{dp_1}{dt}=-\Gamma_{\downarrow}p_1+\Gamma_{\uparrow}(1-p_1)=-\Gamma_1(p_1-p_1^{ss}), \tag{8}$$

$$\Gamma_1\equiv\Gamma_{\downarrow}+\Gamma_{\uparrow}=\frac{1}{T_1},p_1^{ss}=\frac{\Gamma_{\uparrow}}{\Gamma_1},p_1(t)=p_1^{ss}+[p_1(0)-p_1^{ss}]e^{-t/T_1}. \tag{9}$$

A conventional population recovery experiment consequently measures the sum of the upward and downward rates, with its fitted T_1 approaching the inverse downward rate in the low excitation limit. Combining that recovery time with the stationary excited population resolves the individual rates:

$$\Gamma_{\uparrow}=\frac{p_1^{ss}}{T_1},\Gamma_{\downarrow}=\frac{1-p_1^{ss}}{T_1}. \tag{10}$$

This separation is essential when evaluating residual excitation or a deliberately heated microwave environment, because the recovery time and stationary population constrain different aspects of the same two state exchange process. Together, the measurements determine both the overall recovery rate and its directional balance (Ithier et al., 2005).

3 Thermal statistical mechanics fixes the direction balance

Thermal equilibrium weights each energy eigenstate by its Boltzmann factor; choosing zero ground state energy and an excited energy $\hbar\omega_{01}$ gives the canonical state and partition function

$$\rho_{th}=\frac{e^{-\beta\hat{H}_q}}{Z},\beta=\frac{1}{k_BT},Z=1+e^{-\beta\hbar\omega_{01}}, \tag{11}$$

where the energy shifted Hamiltonian $\hat{H}_q=\hbar\omega_{01}|1\rangle\langle 1|$ yields the equilibrium populations and polarization

$$\frac{p_1^{th}}{p_0^{th}}=e^{-\beta\hbar\omega_{01}},p_1^{th}=\frac{1}{1+e^{\beta\hbar\omega_{01}}},\langle\sigma_z\rangle_{th}=-\tanh\left(\frac{\beta\hbar\omega_{01}}{2}\right). \tag{12}$$

The resemblance to a Fermi occupation function arises from normalizing the probabilities of two accessible qubit levels.. By comparison, microwave photons or phonons in the environment obey the Bose distribution, whose occupation enters the rates through the bath coupling.

At stationary equilibrium, equal upward and downward probability currents satisfy $\Gamma\uparrow p_0 = \Gamma\downarrow p_1$, so equation (12) fixes their rate ratio as

$$\frac{\Gamma_\uparrow}{\Gamma_\downarrow}=e^{-\hbar\omega_{01}/(k_BT)}. \tag{13}$$

For a thermal bosonic bath coupled at the transition frequency, the mean occupation $\bar{n}$ and zero temperature spontaneous emission rate γ_0 determine the directed rates through

$$\bar{n}(\omega,T)=\frac{1}{e^{\hbar\omega/(k_BT)}-1},\Gamma_\downarrow=\gamma_0(\bar{n}+1),\Gamma_\uparrow=\gamma_0\bar{n}. \tag{14}$$

The terms proportional to $\bar{n}$ describe stimulated exchange, while the additional downward term describes spontaneous emission. Cooling therefore suppresses upward thermal excitation while leaving a vacuum contribution to downward emission, resulting in

$$\Gamma_1=\gamma_0(2\bar{n}+1),p_1^{th}=\frac{\bar{n}}{2\bar{n}+1},T_{eff}=\frac{\hbar\omega_{01}}{k_B\ln(\Gamma_\downarrow/\Gamma_\uparrow)}. \tag{15}$$

Here, the final expression assigns a transition specific effective temperature when $0 < \Gamma\uparrow < \Gamma\downarrow$, providing a direct link between the measured rate ratio and the thermal population model. In a fabricated device, attenuation, filtering, thermal anchoring, and incoming radiation determine the microwave field at the qubit frequency and thereby mediate the relationship between refrigerator temperature and qubit population. Quasiparticle driven excitation contributes through a further experimentally observed mechanism (Serniak et al., 2018).

At fixed circuit frequency, the excited population determines the mean qubit energy and the heat current entering the qubit:

$$U=\hbar\omega_{01}p_1,\dot{Q}=\hbar\omega_{01}\left[\Gamma_\uparrow(1-p_1)-\Gamma_\downarrow p_1\right]. \tag{16}$$

For a diagonal state with entropy $S=-k_B\left[p_0\ln p_0+p_1\ln p_1\right)$, coupling to a thermal two state bath satisfying equation (13) gives the entropy production

$$\dot{S}-\frac{\dot{Q}}{T}=k_B(a-b)\ln\left(\frac{a}{b}\right)\geq 0\,,a=\Gamma_{\uparrow}p_0\,,b=\Gamma_{\downarrow}p_1\,. \tag{17}$$

Because the logarithm increases with its argument, the product of the current difference and the logarithmic ratio is nonnegative. This expression identifies relaxation as a statistical mechanical process in which unequal forward and reverse probability currents generate entropy as populations approach equilibrium; once a = b, the net heat current vanishes even though individual transitions continue.

4 Microscopic coupling becomes a quantum noise spectrum

Environmental energy exchange enters through the interaction Hamiltonian $\widehat{H}_{int}=\widehat{A}\otimes\widehat{B}$, where Â acts on the circuit, $\hat{\mathrm{B}}$ acts on the bath, and their product has units of energy. Considering bath eigenstates $|\mu\rangle$ and $|\nu\rangle$ with energies ϵ_μ and ϵ_ν, together with initial occupation probabilities P_μ, perturbation theory gives the transition rate

$$\Gamma_{i\to f}=\frac{2\pi}{\hbar}\left|A_{fi}\right)^2\sum_{\mu,\nu}P_\mu\left|B_{\nu\mu}\right)^2\delta\left(E_f-E_i+\epsilon_\nu-\epsilon_\mu\right). \tag{18}$$

Fermi's golden rule applied to the combined circuit and bath expresses energy conservation through the delta function and the strength of each state coupling through the squared matrix element. A constant rate description follows when the bath spectrum is smooth and its correlation time is short relative to the population evolution.

The bath correlation function defines a two sided, unsymmetrized spectrum in angular frequency:

$$S_{BB}(\omega)=\int_{-\infty}^{\infty}dt\,e^{i\omega t}\left\langle\delta\widehat{B}(t)\delta\widehat{B}(0)\right\rangle_B,\delta\widehat{B}=\widehat{B}-\left\langle\widehat{B}\right\rangle_B. \tag{19}$$

Inserting the bath energy eigenstates into equation (19) associates positive frequency weight with absorption of energy by the bath, so defining $\omega_{if}=\left(E_i-E_f\right)/\hbar$ converts equation (18) into

$$\Gamma_{i\to f}=\frac{\left|A_{fi}\right)^2}{\hbar^2}S_{BB}\left(\omega_{if}\right),\Gamma_{\downarrow}=\frac{\left|A_{01}\right)^2}{\hbar^2}S_{BB}\left(+\omega_{01}\right),\Gamma_{\uparrow}=\frac{\left|A_{01}\right)^2}{\hbar^2}S_{BB}\left(-\omega_{01}\right). \tag{20}$$

Because the spectrum has units B²·s, multiplying it by the squared circuit matrix element and dividing by ħ² produces a rate in s⁻¹. Its spectral weight combines the bath mode density with the occupied coupling strengths, whose thermal weighting gives the Kubo Martin Schwinger detailed balance relation

$$S_{BB}(-\omega)=e^{-\beta\hbar\omega}S_{BB}(+\omega),\omega>0. \tag{21}$$

Substituting equation (21) into the directed rates of equation (20) recovers equation (13), identifying the positive and negative frequency asymmetry as the spectral expression of thermal detailed balance (Clerk et al., 2010).

For an equilibrium voltage environment with impedance $Z_{env}(\omega)$, the fluctuation dissipation relation takes the following form in the convention of equation (19):

$$S_{VV}(\omega)=\frac{2\hbar\omega\,\mathrm{Re}\,Z_{env}(\omega)}{1-e^{-\beta\hbar\omega}},\ \overline{S}_{VV}(\omega)=\hbar\omega\coth\left(\frac{\beta\hbar\omega}{2}\right)\mathrm{Re}\,Z_{env}(\omega). \tag{22}$$

The bar denotes the average of $S_{VV}(\omega)$ and $S_{VV}(-\omega)$, whose high temperature limit is the symmetrized two sided spectrum $2k_BT\,\mathrm{Re}\,Z_{env}$. For a resistor R, the corresponding one sided classical voltage spectrum is $4k_BTR$, with the factor of two arising from the change in spectral convention. Whereas symmetrization combines emission and absorption, the separate unsymmetrized values determine the two directed rates, preserving the energy transfer direction when circuit dissipation is related to statistical fluctuations.

For capacitive coupling, a voltage fluctuation produces the interaction $\widehat{H}_{int}=2e\eta\hat{n}\delta V$, where η is the effective capacitive voltage division factor, giving the downward rate

$$\Gamma_{\downarrow}^{(V)}=\frac{(2e\eta)^2}{\hbar^2}|\langle 0|\hat{n}|1\rangle|^2 S_{VV}(\omega_{01}),|\langle 0|\hat{n}|1\rangle\simeq\left(\frac{E_J}{32E_C}\right)^{1/4}. \tag{23}$$

The final matrix element uses the harmonic transmon approximation, while the equivalent flux coupling follows from $\widehat{H}_{int}=-\hat{I}\delta\Phi$ and the current matrix element. When the circuit parameters depart from that approximation, diagonalizing the actual Hamiltonian supplies the appropriate matrix elements for the same spectral rate calculation (Koch et al., 2007).

5 The master equation connects populations and coherence

While a two state rate equation tracks energy populations, a density matrix master equation also describes the complex phase relations between the states. For stationary, weak environmental coupling with resolved transition frequencies, the Gorini Kossakowski Sudarshan Lindblad form gives trace preserving, completely positive evolution (Gorini, Kossakowski and Sudarshan, 1976; Lindblad, 1976):

$$\dot{\rho}=-\frac{i}{\hbar}\left[\widehat{H}_q,\rho\right)+\Gamma_{\downarrow}D\left[\hat{\sigma}_-\right)\rho+\Gamma_{\uparrow}D\left[\hat{\sigma}_+\right)\rho+\frac{\Gamma_\phi}{2}D\left[\hat{\sigma}_z\right)\rho. \tag{24}$$

$$D\left[\hat{L}\right)\rho=\hat{L}\rho\hat{L}^\dagger-\frac{1}{2}\{\hat{L}^\dagger\hat{L},\rho\},\hat{\sigma}_-=|0\rangle\langle 1|,\hat{\sigma}_+=|1\rangle\langle 0|. \tag{25}$$

The commutator generates coherent motion, while the first two dissipators describe downward and upward population transfer and the final dissipator introduces pure dephasing at rate Γ_ϕ. Including a factor of one half before the $\hat{\sigma}_z$ dissipator makes Γ_ϕ the off diagonal decay rate contributed by that term, corresponding to the simulation collapse operators $\sqrt{\Gamma_{\downarrow}}\hat{\sigma}_-$, $\sqrt{\Gamma_{\uparrow}}\hat{\sigma}_+$, and $\sqrt{\Gamma_\phi/2}\hat{\sigma}_z$.

Evaluating the population and coherence matrix elements of equation (24) gives

$$\dot{\rho}_{11}=-\Gamma_1\left(\rho_{11}-p_1^{ss}\right),\dot{\rho}_{10}=-\left(i\omega_{01}+\Gamma_2\right)\rho_{10},\Gamma_2=\frac{\Gamma_1}{2}+\Gamma_\phi. \tag{26}$$

For exponential population recovery and coherence decay, these equations imply

$$\frac{1}{T_2}=\frac{1}{2T_1}+\frac{1}{T_\phi},\rho_{10}(t)=\rho_{10}(0)e^{-i\omega_{01}t}e^{-t/T_2},T_2\le 2T_1. \tag{27}$$

The factor of two reflects the dependence of population on squared amplitude, whereas off diagonal coherence depends on an amplitude product. Energy relaxation therefore damps coherence at half the population recovery rate, to which pure dephasing adds phase uncertainty without transferring population between the two energy levels.

Combining the population recovery with the complex coherence gives the full solution in the ordered basis $\{|0\rangle, |1\rangle\}$:

$$\rho(t)=\begin{pmatrix} 1-p_1(t) & \rho_{01}(0)e^{+i\omega_{01}t-t/T_2} \\ \rho_{10}(0)e^{-i\omega_{01}t-t/T_2} & p_1(t) \end{pmatrix}. \tag{28}$$

By separating energy recovery from phase coherence, equation (28) provides a direct numerical check on a two level master equation solver. Within this model, T_2 denotes an exponential transverse decay time; because Ramsey and echo experiments weight environmental fluctuations differently, their measured envelopes and fitted decay times carry the corresponding pulse sequence label (Ithier et al., 2005).

6 Frequency fluctuations produce dephasing

Environmental coupling contains transverse components that connect different energy eigenstates and drive transitions, together with longitudinal components that shift the energy splitting and change the accumulated phase between experimental shots. Writing the longitudinal frequency fluctuation as $\delta\omega(t)$ gives the Hamiltonian and pure dephasing function

$$\widehat{H}(t)=\frac{\hbar}{2}\left[\omega_{01}+\delta\omega(t)\right)\hat{\sigma}_z, W(t)=\left\langle e^{-i\int_0^t \delta\omega(s)ds}\right\rangle. \tag{29}$$

Multiplying W(t) by $\exp[-t/(2T_1)]$ gives the normalized coherence, while zero mean Gaussian classical frequency fluctuations yield $W = \exp(-\chi)$ through the characteristic function

$$\begin{aligned} \chi(t) \quad & \frac{1}{2}\langle\phi(t)^2\rangle=\frac{1}{2}\int_{-\infty}^{\infty}\frac{d\omega}{2\pi}S_{\delta\omega}(\omega)\left|Y(\omega,t)\right)^2, \\ Y(\omega,t) \quad & \int_0^t y(s)e^{i\omega s}ds. \end{aligned} \tag{30}$$

During free Ramsey evolution y(s) = 1, whereas an ideal refocusing π pulse reverses the phase accumulation and changes its sign, producing the respective filter weights

$$\left|Y_R\right)^2=\frac{4\sin^2(\omega t/2)}{\omega^2},\left|Y_E\right)^2=\frac{16\sin^4(\omega t/4)}{\omega^2}. \tag{31}$$

For a single instantaneous π pulse at t/2, the echo filter suppresses low frequency noise and extends coherence when slow frequency drift dominates. Additional pulses reshape that frequency weighting, enabling experimental reconstruction of selected parts of the noise spectrum, as demonstrated by Bylander et al. (2011).

White frequency noise with the two sided value $S_{\delta\omega}(\omega)=S_0$ gives $\chi = S_0 t/2$ and $\Gamma_\phi = S_0/2$, whereas quasistatic Gaussian fluctuations with variance σ_ω^2 produce the envelope

$$W_R(t)=e^{-\sigma_\omega^2 t^2/2}, T_{\phi,G}=\frac{\sqrt{2}}{\sigma_\omega}. \tag{32}$$

An exponential lifetime and a Gaussian decay scale consequently describe different statistical processes, with the complete Ramsey envelope obtained by multiplying the pure dephasing function by the relaxation factor. A fitted Ramsey time T_2^* or echo time T_2E becomes a physically interpretable specification when its associated envelope form is stated.

For a small fluctuation in an electrical control parameter λ, the circuit sensitivity converts parameter noise into frequency noise according to

$$\delta\omega \simeq \frac{\partial\omega_{01}}{\partial\lambda}\delta\lambda, S_{\delta\omega}(\omega)=\left|\frac{\partial\omega_{01}}{\partial\lambda}\right|^2 S_\lambda(\omega). \tag{33}$$

Equation (33) separates the noise amplitude from the circuit sensitivity, showing that a bias point with $\partial\omega_{01}/\partial\lambda = 0$ suppresses first order coupling to that parameter. The next contribution follows from the second derivative, which establishes the mathematical basis of a coherence sweet spot and connects the control landscape to phase stability (Vion et al., 2002).

7 Material and electromagnetic design determine loss channels

Circuit geometry determines the local fields, while the material response determines how much energy those fields dissipate. By combining mode frequencies and field distributions from finite element electromagnetic simulations with state matrix elements from circuit quantization, physical dimensions and material parameters are translated into transition rates. Energy participation quantization provides a systematic formulation of this connection for distributed Josephson circuits (Minev et al., 2021).

Dielectric participation and quality factor

For a weakly lossy mode, the fraction of electric field energy in region j defines its participation and contribution to dielectric loss:

$$p_j=\frac{\int_{V_j}\epsilon_j'|E(r)|^2 dV}{\int_{V_{all}}\epsilon'(r)|E(r)|^2 dV}, Q_{diel}^{-1}=\sum_j p_j\tan\delta_j, \Gamma_{1,diel}\simeq\omega_{01}\sum_j p_j\tan\delta_j. \tag{34}$$

The common energy prefactor cancels between the integrals, whose denominator contains the total electric energy as opposed to the combined electric and magnetic mode energy, thereby avoiding a factor of two in the participation definition. In the low excitation, weak dielectric loss regime, the loss tangent $\tan\delta_j=\epsilon_j''/\epsilon_j'$ measures the dissipative material response, so field concentration at a metal oxide interface and its thin dissipative layer jointly determine the contribution $p_j\tan\delta_j$ (Wang et al., 2015).

An energy decay time T_1 corresponds to the quality factor $Q_1 = \omega_{01}T_1$, allowing measured lifetimes to be compared with electromagnetic loss calculations. Because dielectric defects with two accessible microscopic configurations interact with microwave electric fields and produce temperature and power

dependent loss, measurements at the device temperature and near single photon occupation supply the relevant qubit loss parameters. Following the identification of dielectric loss in Josephson qubits by Martinis et al. (2005), participation analysis connects that material response to device geometry by combining a calculated electric energy fraction with a measured loss tangent.

Purcell relaxation through the readout circuit

Coupling to a resonator adds a small resonator component to the qubit eigenstate, opening a path for qubit energy to escape through the resonator into the transmission line. In the dispersive regime $|\Delta| \gg g, \kappa$, with detuning $\Delta = \omega_{01} - \omega_r$, this path gives the approximate cold environment Purcell rate

$$\Gamma_P \simeq \left(\frac{g}{\Delta}\right)^2 \kappa, \kappa = \frac{\omega_r}{Q_r}, T_{1,P} = \Gamma_P^{-1}. \tag{35}$$

Here, κ describes resonator energy decay, so the field amplitude decays at $\kappa/2$, while g/Δ remains dimensionless when both parameters use the same frequency convention. For a thermal resonator bath, the same approximation gives $\Gamma_{\downarrow,P} = \Gamma_P(\bar{n}+1)$ and $\Gamma_{\uparrow,P} = \Gamma_P \bar{n}$, separating stimulated exchange from spontaneous emission. A Purcell filter suppresses this decay path by modifying the admittance at the qubit frequency while preserving a transmission band for readout (Blais et al., 2021).

Quasiparticle tunneling

Unpaired quasiparticle excitations alongside the superconducting condensate provide a tunneling mechanism across the Josephson element that exchanges energy with the qubit. A standard low energy junction model expresses this coupling through the phase matrix element

$$\Gamma_{i\to f}^{qp} = \left|\left\langle f \left| \sin\frac{\hat{\varphi}}{2} \right| i \right\rangle\right|^2 S_{qp}(\omega_{if}). \tag{36}$$

The quasiparticle spectrum S_{qp} includes the tunneling prefactors and therefore has units s^{-1}; for a cold distribution with characteristic excitation energies near the superconducting gap, the small phase transmon approximation at zero phase bias reduces its downward rate to

$$\Gamma_{\downarrow}^{qp} \simeq \frac{\omega_{01}}{\pi} x_{qp} \sqrt{\frac{2\Delta_{sc}}{\hbar\omega_{01}}}, x_{qp} = \frac{n_{qp}}{2\nu_0 \Delta_{sc}}. \tag{37}$$

The superconducting gap Δ_{sc} differs from the resonator detuning Δ, while the normalized quasiparticle concentration uses the number density n_{qp} and the normal density of states ν_0 per spin, per energy, and per volume. At low temperature, a thermal distribution gives $x_{qp} \simeq \sqrt{\frac{2\pi k_B T}{\Delta_{sc}}} \exp\left[-\Delta_{sc}/(k_B T)\right]$, although the general spectral expression also describes distributions produced by radiation and nonequilibrium energy injection (Catelani et al., 2011). Measurements of excitation and relaxation that resolve quasiparticle parity changes identify this contribution experimentally (Serniak et al., 2018).

Combining channels into a rate budget

Independent weak environmental channels provide alternative paths for each transition, so their directed rates add according to

$$\Gamma_{\downarrow}=\sum_{a}\Gamma_{\downarrow,a},\Gamma_{\uparrow}=\sum_{a}\Gamma_{\uparrow,a},\frac{1}{T_1}=\sum_{a}\left(\Gamma_{\downarrow,a}+\Gamma_{\uparrow,a}\right). \tag{38}$$

A rate budget assigns each material region or external port a contribution in s^{-1}, with conductive loss, radiation, seams, and additional package modes entering through dissipated power or an equivalent environmental spectrum. For weakly damped, nearly harmonic electrical motion, $P_{loss}/U=\omega/Q$ gives the energy decay rate, which maps onto the measured downward dominated T_1 in the low temperature qubit limit.

A single strongly coupled microscopic defect requires an explicit place in the system Hamiltonian because coherent exchange with that defect produces an avoided crossing or an oscillatory decay trace. Including its state in the numerical model preserves this behavior, whereas a sum of independent decay rates describes alternative weak loss paths.

8 From measured counts to rates and uncertainty

A T_1 experiment measures population recovery by preparing $|1\rangle$, waiting for a specified delay, and repeating the state measurement, with the delay dependence described by equation (9). Readout assignment maps the physical population to the observed probability and the corresponding count likelihood through

$$q(t)=\epsilon_0+\left(1-\epsilon_0-\epsilon_1\right)p_1(t),L=\prod_{k}\binom{N_k}{K_k}q_k^{K_k}\left(1-q_k\right)^{N_k-K_k}. \tag{39}$$

At each delay t_k, the probability $q_k=q\left(t_k\right)$ generates K_k excited reports among N_k trials, with ε_0 = Pr(report 1 | state 0) and ε_1 = Pr(report 0 | state 1) specifying the assignment errors. Maximizing the binomial likelihood fits the population model directly to those counts, while initialization errors enter through the fitted or calibrated $p_1(0)$. Keeping initialization and readout parameters explicit prevents an observed population offset from being absorbed incorrectly into the fitted lifetime.

For a calibrated probability estimate $\hat{p}=K/N$, the approximate binomial standard error $\sqrt{\hat{p}(1-\hat{p})/N}$ quantifies shot uncertainty, while a local one parameter Fisher information calculation relates the lifetime precision to the chosen delays:

$$I\left(T_1\right)=\sum_{k}\frac{N_k}{q_k\left(1-q_k\right)}\left(\frac{\partial q_k}{\partial T_1}\right)^2,Var\left(\hat{T}_1\right)\geq I\left(T_1\right)^{-1}. \tag{40}$$

The Cramér Rao bound applies to a regular unbiased estimator in this one parameter model, while joint estimation of offsets, amplitudes, or readout errors requires the full Fisher information matrix. Repeating the lifetime fit over time measures drift in addition to the shot uncertainty within each trace, so reporting the central value with the acquisition interval and spread distinguishes these two sources of variation.

Continuous monitoring estimates the same rates from individually resolved jumps by dividing downward transition counts by the total dwell time in $|1\rangle$ and upward counts by the total dwell time in $|0\rangle$. For constant rates, the likelihood gives the estimators

$$\widehat{\Gamma}_{\downarrow}=\frac{N_{\downarrow}}{\tau_1},\widehat{\Gamma}_{\uparrow}=\frac{N_{\uparrow}}{\tau_0}. \tag{41}$$

Because the exposure times τ_1 and τ_0 are conditioned on the occupied state, timing resolution and state assignment enter the measurement model used to extract them. The dwell time estimators in equation (41) and the ensemble recovery fit in equation (39) therefore estimate the same physical rates through different observables.

A Ramsey experiment prepares a superposition and, after free evolution, converts the accumulated phase into population with a second $\pi/2$ pulse, giving the fitting form $q_R(t)=A+Be^{-t/(2T_1)}W_R(t)\cos(\delta t+\phi_0)$, where δ is the drive detuning. Adding a refocusing pulse produces an echo experiment and replaces W_R with W_E, allowing the measured trace to identify both the coherence envelope and the frequency offset (Ithier et al., 2005).

9 Gate operations and the coherence budget

In the rotating frame and rotating wave approximation, a phase controlled resonant microwave pulse drives coherent rotation through the Hamiltonian

$$\widehat{H}_d(t)=\frac{\hbar\Omega_R(t)}{2}\left[\cos\theta\,\hat{\sigma}_x+\sin\theta\,\hat{\sigma}_y\right],\vartheta=\int_0^{t_g}\Omega_R(t)\,dt. \tag{42}$$

The drive phase θ selects the rotation axis, while the integrated Rabi angular frequency sets the rotation angle ϑ. For a constant resonant drive starting in $|0\rangle$, the ideal excited population follows $\sin^2(\Omega_R t/2)$, giving a π pulse duration $t_\pi=\pi/\Omega_R$. Replacing $\widehat{H}_q$ in equation (24) with the driven Hamiltonian and integrating the master equation then incorporates dissipation into the controlled evolution.

Rabi frequency describes controlled coherent motion, whereas a stochastic transition rate describes incoherent energy exchange, with the ratios t_g/T_1 and t_g/T_2 measuring the operation duration against the relevant decay scales. Although shorter pulses reduce exposure to dissipation, finite anharmonicity and pulse spectral width govern unwanted population of higher levels, so circuit parameters and waveform design jointly determine useful gate speed (Krantz et al., 2019).

For an idle lasting t_g with stationary exponential relaxation and dephasing, removing the known free evolution phase gives the average fidelity relative to the identity operation as

$$F_{avg}(t_g)=\frac{1}{2}+\frac{2e^{-t_g/T_2}+e^{-t_g/T_1}}{6},r_{idle}\equiv 1-F_{avg}. \tag{43}$$

Averaging the channel fidelity over pure states on the Bloch sphere removes the contribution from the affine shift toward thermal equilibrium, leaving the exponential decay terms whose short time expansion gives

$$r_{idle} \simeq \frac{t_g}{6}\left(\frac{2}{T_2}+\frac{1}{T_1}\right)=\frac{t_g}{3T_1}+\frac{t_g}{3T_\phi}. \tag{44}$$

Equation (43) gives the exact idle channel result within this model, and equation (44) supplies its short time approximation for comparing operation durations with coherence scales. Evaluating a driven gate instead uses its actual Hamiltonian and noise channel, because average gate fidelity measures performance across input states, while 1 − exp(−Γ↓t) gives the downward jump probability for an initially excited state. These observables consequently answer different experimental questions even when they depend on the same transition rates (Nielsen, 2002).

In a multilevel circuit, computational leakage measures occupation outside $\{|0\rangle, |1\rangle\}$, separating escape from the computational subspace from relaxation between its two states:

$$P_{leak}(t)=1-Tr\left[\widehat{\Pi}_{comp}\rho(t)\right], \widehat{\Pi}_{comp}=|0\rangle\langle 0|+|1\rangle\langle 1|. \tag{45}$$

While T_1 and T_2 describe decay processes, gate benchmarking measures the implemented operation through sequences whose observed errors reflect its executed dynamics. Randomized benchmarking estimates average errors from gate sequences, and interleaved benchmarking assesses a selected operation through comparison sequences (Magesan et al., 2012). At the processor level, quantum volume uses random circuits of matched width and depth, making connectivity, compilation, simultaneous operation, and readout part of the system measurement (Cross et al., 2019).

10 Worked calculations for a 5 GHz qubit

For an illustrative two state device with f_{01} = 5 GHz, T_1 = 100 μs, and exponential pure dephasing time T_ϕ = 150 μs, the constants h = 6.62607015 × 10^{-34} J·s and k_B = 1.380649 × 10^{-23} J·K^{-1} give

$$\frac{hf_{01}}{k_B}=0.23996\,K, hf_{01}=3.3130\times10^{-24}\,J=20.678\,\mu eV, T_2=85.714\,\mu s. \tag{46}$$

Equation (12) gives the thermal populations in Table 1, whose temperature dependence follows from the approximately 240 mK thermal energy scale of a 5 GHz photon. The exponential dependence makes the occupation small at 20 mK and appreciable at 100 mK, directly connecting microwave bath temperature to residual excited population.

Table 1 Thermal occupation for a 5 GHz two state qubit

Bath temperature (mK)	Energy ratio hf_{01} / kBT	Bosonic occupation $\bar{n}$	Excited population (%)
20	11.998	6.15589e-06	0.000615581
50	4.799	0.00830437	0.81687
100	2.400	0.0998103	8.32016
200	1.200	0.431129	23.1509

At 50 mK, the stationary thermal population $p_1 = 0.00817$ and a measured recovery time of 100 μs give $\Gamma\uparrow = 81.7\ s^{-1}$ and $\Gamma\downarrow = 9.918 \times 10^3\ s^{-1}$ through equation (10). The corresponding mean first downward waiting time is 100.82 μs, whereas the population recovery time remains 100 μs because it includes both transition directions.

With negligible upward excitation, an initially excited state has a downward jump probability of $1 - \exp(-0.01) = 0.009950$, or 0.9950%, over 1 μs; reducing the interval to 20 ns lowers that probability to 0.019998%. These values quantify the decay exposure over each operation interval and show why gate duration must be evaluated numerically alongside the lifetime.

For the same T_1 and T_ϕ, equation (43) gives idle average infidelities of 1.111×10^{-4} at 20 ns and 5.525×10^{-3} at 1 μs, which Table 2 compares with the distinct population and coherence observables.

Table 2 Decay exposure for T_1 of 100 μs and T_φ of 150 μs

Dimensionless quantity	**20 ns**	**1 μs**
Duration relative to T_1	0.0002	0.01
Downward jump probability	0.00019998	0.009950166
Population survival	0.9998	0.9900498
Normalized coherence	0.9997667	0.9884011
Idle average infidelity	0.0001110987	0.005524653

For an ideal low excitation ensemble of 1,000 preparations at $t = T_1$, the binomial distribution predicts 367.9 excited outcomes with a standard deviation of 15.25 counts. This corresponds to approximately 1.53 percentage points of population uncertainty from shot statistics alone, which decreases in proportion to $N^{-1/2}$ as the number of repetitions grows.

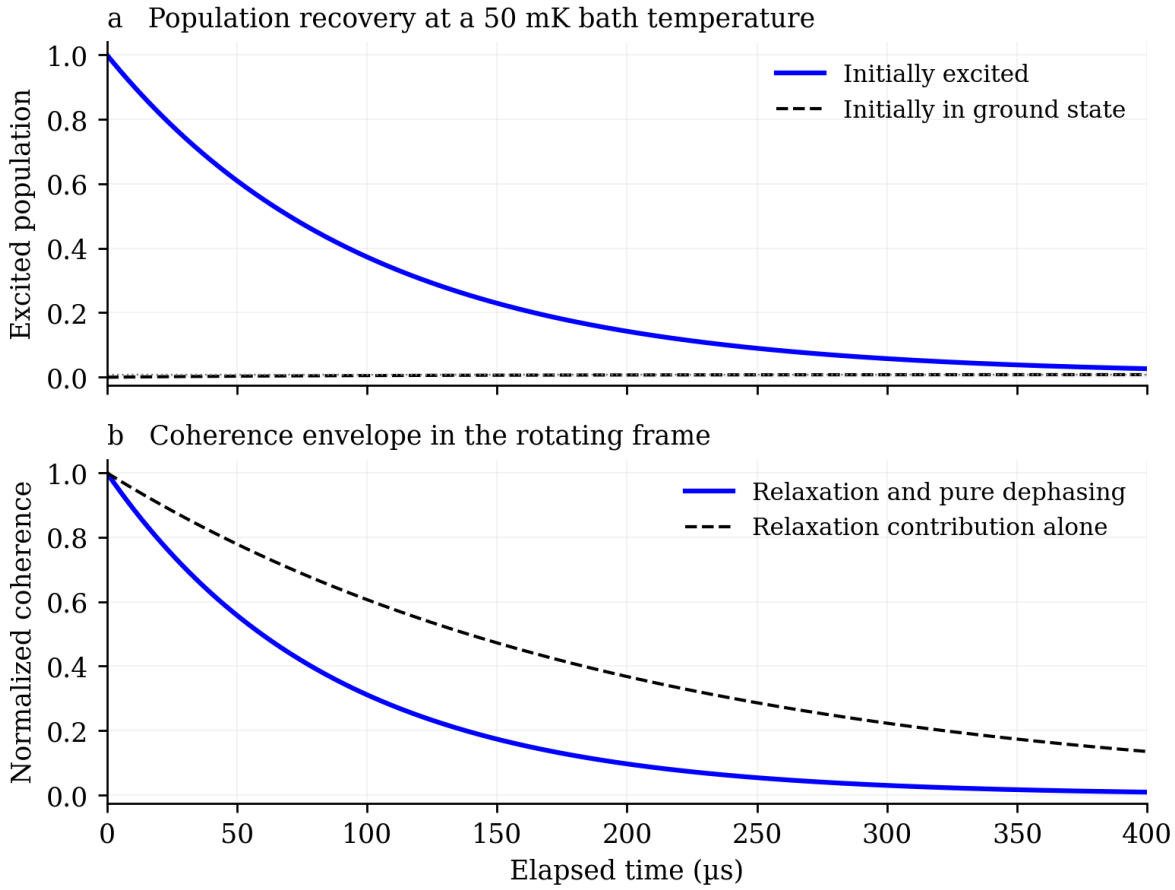


Figure 1 Population recovery and coherence for f_{01} = 5 GHz, T_1 = 100 μs, and $T\varphi$ = 150 μs. Panel a shows recovery toward a stationary excited population of 0.8169%, while panel b shows coherence after removal of the deterministic free precession phase.

The same device parameters translate a material loss budget into a design target, since f_{01} = 5 GHz and T_1 = 100 μs imply $Q_1 = 3.142 \times 10^6$ and $Q_1^{-1} = 3.183 \times 10^{-7}$. If one dielectric region with $\tan\delta = 10^{-3}$ accounts for that loss, equation (34) requires $p = 3.183 \times 10^{-4}$, or 0.03183% of the electric energy. The product $p \tan\delta$ thereby connects the measured material response to the field participation calculated from device geometry.

For a cold readout resonator with $g/(2\pi)$ = 100 MHz, $|\Delta|/(2\pi)$ = 1 GHz, and $\kappa/(2\pi)$ = 1 MHz, equation (35) gives $\Gamma_P = 6.283 \times 10^4\ s^{-1}$ and T_1P = 15.92 μs. Holding that coupling ratio fixed, a Purcell contribution corresponding to at least 100 μs requires an effective $\kappa/(2\pi)$ of at most 0.1592 MHz at the qubit frequency, providing a quantitative target for the frequency dependent suppression engineered by a filter.

Using Δ_{sc} = 180 μeV and $x_{qp} = 10^{-7}$ in the cold distribution and transmon approximation of equation (37) gives a downward rate of approximately $4.17 \times 10^3\ s^{-1}$, corresponding to a quasiparticle limited lifetime near 240 μs. Combining this contribution with other independent loss channels increases the total rate according to equation (38), linking the quasiparticle concentration to the overall coherence budget.

11 A numerical model connects fabrication to measurement

A numerical device model begins by extracting C_Σ, field participation, resonator frequencies, and external port coupling from a capacitance or eigenmode calculation. Incorporating the Josephson energy then produces a nonlinear circuit Hamiltonian whose diagonalization gives transition frequencies and state matrix elements, with the basis enlarged until those quantities and the predicted rates converge. Energy participation quantization supports this connection between spatial field solutions and quantum mode parameters (Minev et al., 2021).

Each relevant coupling coordinate then receives an environmental spectrum, with measured dielectric loss, extracted impedance, and quasiparticle distributions evaluated at the applicable positive and negative transition frequencies. The directed rates from equation (20) enter the master equation, while low frequency longitudinal spectra enter the coherence filter integral, so electromagnetic design and noise calibration supply complementary inputs to the same dynamical calculation.

Extending the calculation to M levels gives the population dynamics and normalization condition

$$\frac{d\,p_i}{d\,t}=\sum_{j\neq i}\left(\Gamma_{j\to i}\,p_j-\Gamma_{i\to j}\,p_i\right),\ \sum_{i=0}^{M-1}p_i=1. \tag{47}$$

The corresponding density matrix simulation combines collapse operators $\sqrt{\Gamma_{i\to j}}|j\rangle\langle i|$ with a Hamiltonian containing the applied pulse, using enough transmon levels to resolve pulse induced leakage and the $|1\rangle \leftrightarrow |2\rangle$ transition. Narrow resonators or strongly coupled defects receive explicit states when their internal dynamics shape the measured trace, preserving coherent exchange that would be lost in an independent rate description.

Comparisons with hardware preserve the measured observable by fitting simulated population recovery with the experimental T_1 model and by including drive detuning and the appropriate coherence filter in simulated Ramsey traces. Comparing the simulated channel with the intended unitary then determines numerical gate fidelity, allowing changes in geometry, material loss, or pulse shape to be traced to the same specifications reported by the experiment.

For uncertain device parameters collected in a vector θ with covariance matrix C_θ, first order propagation gives the rate variance

$$Var(\Gamma)\simeq\left(\nabla_\theta\Gamma\right)^T C_\theta\left(\nabla_\theta\Gamma\right). \tag{48}$$

The gradient and covariance identify whether geometric dimensions, uncertain loss tangents, or measured environmental spectra dominate the predicted rate uncertainty. When the dependence is strongly nonlinear, sampling the parameters through the complete model supplies the resulting distributions, allowing the calculation to predict both a loss contribution and its sensitivity to fabrication or control changes.

12 Why these quantities became hardware specifications

The relaxation framework introduced by Bloch's 1946 treatment of nuclear induction linked longitudinal and transverse decay to measurable dynamics, establishing concepts that later informed pulsed spin resonance and quantum control. The independently developed master equation structures of Gorini, Kossakowski and Sudarshan, and of Lindblad in 1976 extended that description to general quantum

evolution under environmental coupling. Superconducting circuits inherited these tools because their quantized electrical states undergo the same population exchange and phase decay processes (Bloch, 1946; Gorini, Kossakowski and Sudarshan, 1976; Lindblad, 1976).

Following the demonstration of coherent control in a single Cooper pair box by Nakamura, Pashkin and Tsai in 1999, work by Vion and colleagues in 2002 developed controlled electrical circuit states with reduced first order sensitivity to fluctuations, while Koch and colleagues formalized the transmon design in 2007. Together with contributions by Devoret, Schoelkopf, Martinis, and their collaborators to circuit design, measurement, and the identification of physical loss mechanisms, these developments connected controllable nonlinear circuits to quantitative noise engineering (Nakamura, Pashkin and Tsai, 1999; Vion et al., 2002; Koch et al., 2007; Martinis et al., 2005).

T_1 and coherence times became common hardware specifications because they link measured decay traces to physical device models developed through this experimental and theoretical work. Identifying the experiment, operating frequency, pulse sequence, and statistical meaning of each fitted parameter carries that physical interpretation into a hardware comparison, as summarized in Table 3.

Table 3 Hardware quantities and their defining measurements with unit 1 denoting a dimensionless value

Quantity	Physical meaning	Measurement	Units
f_{01} and $\alpha / 2\pi$	Transition frequency and adjacent transition separation	Spectroscopy	Hz
$\Gamma\downarrow$	Conditional downward transition rate	State resolved dwell times or T_1 plus stationary population	s^{-1}
$\Gamma\uparrow$	Conditional upward transition rate	State resolved dwell times or T_1 plus stationary population	s^{-1}
T_1	Population recovery time to the stationary value	Excited state preparation and delayed measurement	s
T_2^*	Ramsey coherence decay scale with stated envelope	Two $\pi/2$ pulses separated by free evolution	s
T_2E	Echo coherence decay scale with stated envelope	Ramsey sequence with a refocusing π pulse	s
Pure dephasing	Pure dephasing time or normalized coherence function	Joint population and coherence model	s or 1
p_1 in steady state	Residual excited population	Calibrated state population measurement	1
Gate timing	Operation duration and coherent rotation frequency	Pulse timing and Rabi measurement	s and Hz
Average gate error	Error of the implemented channel relative to the target	Gate benchmarking with stated protocol	1
Computational leakage	Population outside the computational subspace	Measurement resolving higher energy levels	1
Readout assignment	Conditional reporting probabilities for prepared states	Ground and excited state calibration	1

For an audience familiar with magnetic data storage, statistical switching and dwell times provide a direct connection to the characterization of a physical information bearing element. Quantum information

additionally depends on phase coherence between the two states, so longitudinal relaxation changes their populations while transverse decay reduces interference associated with their relative phase. Because different spectral components of the electromagnetic environment govern these effects, a complete hardware description connects energy exchange and dephasing to the performance of executed gates.

The engineering objective is to accumulate the required coherent rotation while keeping environmental error sufficiently small over the operation interval. Circuit Hamiltonians define that intended motion, while thermal statistical mechanics and noise spectra determine the balance and rates of unwanted transitions. Combining these descriptions explains why transition rates and coherence measurements are reported alongside gate duration, readout performance, and computational leakage in superconducting quantum hardware specifications.

References

[1] Blais, A., Grimsmo, A.L., Girvin, S.M. and Wallraff, A. (2021) 'Circuit quantum electrodynamics', *Reviews of Modern Physics*, 93, 025005. https://doi.org/10.1103/RevModPhys.93.025005

[2] Bloch, F. (1946) 'Nuclear induction', *Physical Review*, 70, pp. 460–474. https://doi.org/10.1103/PhysRev.70.460

[3] Bylander, J., Gustavsson, S., Yan, F., Yoshihara, F., Harrabi, K., Fitch, G., Cory, D.G., Nakamura, Y., Tsai, J.S. and Oliver, W.D. (2011) 'Noise spectroscopy through dynamical decoupling with a superconducting flux qubit', *Nature Physics*, 7, pp. 565–570. https://doi.org/10.1038/nphys1994

[4] Catelani, G., Koch, J., Frunzio, L., Schoelkopf, R.J., Devoret, M.H. and Glazman, L.I. (2011) 'Quasiparticle relaxation of superconducting qubits in the presence of flux', *Physical Review Letters*, 106, 077002. https://doi.org/10.1103/PhysRevLett.106.077002

[5] Clerk, A.A., Devoret, M.H., Girvin, S.M., Marquardt, F. and Schoelkopf, R.J. (2010) 'Introduction to quantum noise, measurement, and amplification', *Reviews of Modern Physics*, 82, pp. 1155–1208. https://doi.org/10.1103/RevModPhys.82.1155

[6] Cross, A.W., Bishop, L.S., Sheldon, S., Nation, P.D. and Gambetta, J.M. (2019) 'Validating quantum computers using randomized model circuits', *Physical Review A*, 100, 032328. https://doi.org/10.1103/PhysRevA.100.032328

[7] Gorini, V., Kossakowski, A. and Sudarshan, E.C.G. (1976) 'Completely positive dynamical semigroups of N-level systems', *Journal of Mathematical Physics*, 17, pp. 821–825. https://doi.org/10.1063/1.522979

[8] Ithier, G., Collin, E., Joyez, P., Meeson, P.J., Vion, D., Esteve, D., Chiarello, F., Shnirman, A., Makhlin, Y., Schriefl, J. and Schön, G. (2005) 'Decoherence in a superconducting quantum bit circuit', *Physical Review B*, 72, 134519. https://doi.org/10.1103/PhysRevB.72.134519

[9] Koch, J., Yu, T.M., Gambetta, J., Houck, A.A., Schuster, D.I., Majer, J., Blais, A., Devoret, M.H., Girvin, S.M. and Schoelkopf, R.J. (2007) 'Charge-insensitive qubit design derived from the Cooper pair box', *Physical Review A*, 76, 042319. https://doi.org/10.1103/PhysRevA.76.042319

[10] Krantz, P., Kjaergaard, M., Yan, F., Orlando, T.P., Gustavsson, S. and Oliver, W.D. (2019) 'A quantum engineer's guide to superconducting qubits', *Applied Physics Reviews*, 6, 021318. https://doi.org/10.1063/1.5089550

[11] Lindblad, G. (1976) 'On the generators of quantum dynamical semigroups', *Communications in Mathematical Physics*, 48, pp. 119–130. https://doi.org/10.1007/BF01608499

[12] Magesan, E., Gambetta, J.M., Johnson, B.R., Ryan, C.A., Chow, J.M., Merkel, S.T., da Silva, M.P., Keefe, G.A., Rothwell, M.B., Ohki, T.A., Ketchen, M.B. and Steffen, M. (2012) 'Efficient measurement of quantum gate error by interleaved randomized benchmarking', *Physical Review Letters*, 109, 080505. https://doi.org/10.1103/PhysRevLett.109.080505

[13] Martinis, J.M., Cooper, K.B., McDermott, R., Steffen, M., Ansmann, M., Osborn, K.D., Cicak, K., Oh, S., Pappas, D.P., Simmonds, R.W. and Yu, C.C. (2005) 'Decoherence in Josephson qubits from dielectric loss', *Physical Review Letters*, 95, 210503. https://doi.org/10.1103/PhysRevLett.95.210503

[14] Minev, Z.K., Leghtas, Z., Mundhada, S.O., Christakis, L., Pop, I.M. and Devoret, M.H. (2021) 'Energy-participation quantization of Josephson circuits', *npj Quantum Information*, 7, 131. https://doi.org/10.1038/s41534-021-00461-8

[15] Nakamura, Y., Pashkin, Yu.A. and Tsai, J.S. (1999) 'Coherent control of macroscopic quantum states in a single-Cooper-pair box', *Nature*, 398, pp. 786–788. https://doi.org/10.1038/19718

[16] Nielsen, M.A. (2002) 'A simple formula for the average gate fidelity of a quantum dynamical operation', *Physics Letters A*, 303, pp. 249–252. https://doi.org/10.1016/S0375-9601(02)01272-0

[17] Serniak, K., Hays, M., de Lange, G., Diamond, S., Shankar, S., Burkhart, L.D., Frunzio, L., Houzet, M. and Devoret, M.H. (2018) 'Hot nonequilibrium quasiparticles in transmon qubits', *Physical Review Letters*, 121, 157701. https://doi.org/10.1103/PhysRevLett.121.157701

[18] Vion, D., Aassime, A., Cottet, A., Joyez, P., Pothier, H., Urbina, C., Esteve, D. and Devoret, M.H. (2002) 'Manipulating the quantum state of an electrical circuit', *Science*, 296, pp. 886–889. https://doi.org/10.1126/science.1069372

[19] Wang, C., Axline, C., Gao, Y.Y., Brecht, T., Chu, Y., Frunzio, L., Devoret, M.H. and Schoelkopf, R.J. (2015) 'Surface participation and dielectric loss in superconducting qubits', *Applied Physics Letters*, 107, 162601. https://doi.org/10.1063/1.4934486